

ASHIQ Y

SENIOR SOFTWARE ENGINEER

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PROFESSIONAL SUMMARY

Java backend engineer with 4 years of experience designing and building scalable, distributed backend systems using Java (8–21), Spring Boot, and microservices. Currently building core services for the Netherlands' national bottle-and-can Deposit Return System, processing millions of records daily. Led a 10x improvement in event-log retrieval by migrating 15 years of historical data to Elasticsearch, and modernized legacy Spring Boot applications to Java 21. Comfortable owning features end-to-end, mentoring junior engineers, and collaborating across product, QA, and design to ship reliable, well-tested systems.

WORK EXPERIENCE

Senior Software Engineer — SparkSupport Infotech Pvt Ltd, Kochi, India

Nov 2023 – Present

- Promoted to Senior Software Engineer based on consistent technical contributions, backend ownership, mentoring, and successful delivery of enterprise projects.
- Own the end-to-end backend development lifecycle for multiple enterprise applications — from requirement analysis and design through deployment and production support.
- Build and maintain backend services for the Netherlands' Statiegeld Nederland deposit return system, processing millions of records daily for bottle and can return flows.
- Design reusable backend components and shared libraries — including contributions to the shared HAWK libraries — reducing duplicate implementation across multiple projects.
- Migrated 15 years of historical event-log data from MySQL to Elasticsearch, achieving a 10x improvement in retrieval performance and enabling faster incident investigation.
- Optimize complex QueryDSL and SQL queries for transaction reporting and dashboard APIs, and improved backend performance through Redis-based and local caching for frequently accessed data.
- Led modernization of multiple enterprise applications — upgrading Java (8/11 to 21), Spring Boot (2.x/3.x to 4.0), and Spring Security to 6.x using OpenRewrite — while maintaining production stability across international deployments.
- Implement secure authentication and authorization using JWT and OAuth 2.0, and Active Directory integrations.
- Design and maintain RabbitMQ-based asynchronous processing workflows for scalable, high-volume transaction handling across distributed deployments.
- Design and implement REST APIs for transaction processing, machine management, reporting, and configuration management, collaborating closely with frontend teams on API contracts.
- Integrated AWS S3 and Azure Blob Storage for cloud-based storage.
- Sole developer for RVMCOL, built from scratch and now deployed across multiple countries; concurrently develop and deliver features across 5+ client projects (SRN, SNL, OCM, HAWK, RVM Connector).
- Diagnose and resolve production issues across Docker, Docker Swarm, Kubernetes, VPN connectivity, and distributed environments; member of the production release team handling merges, deployment coordination, hotfixes, and post-release monitoring.
- Fixed SonarQube code quality issues across multiple repositories and maintained JUnit/Mockito unit and integration tests to increase reliability.
- Onboarded 8+ new developers and mentored freshers and interns, streamlining the onboarding process and improving team productivity.
- Maintained a 95%+ on-time sprint delivery rate through close collaboration with QA and product teams across international deployments.

Software Engineer – PAT — Cognizant, Bangalore, India

Jul 2022 – Aug 2023

- Developed and maintained scalable RESTful APIs using Java 8, Spring Boot, Hibernate/JPA, and a microservices architecture.
- Managed relational databases (MySQL), ensuring data integrity and optimal query performance.
- Wrote unit and integration tests with JUnit to validate functionality and maintain high test coverage.
- Refactored and optimized code, removing duplicate logic and improving reusability, maintainability, and application performance.

TECHNICAL SKILLS

- **Languages:** Java 8, 11, 17, 21
- **Frameworks & ORM:** Spring Boot, Spring Security, Hibernate, JPA, QueryDSL, JasperReports
- **Architecture:** Microservices, REST APIs, Monolith-to-Microservices Migration
- **Databases & Search:** MySQL, MariaDB, MongoDB, Elasticsearch, Kibana
- **Messaging & Caching:** RabbitMQ, Redis, Consul
- **Security:** OAuth 2.0, JWT, Active Directory, OpenVPN
- **Testing & Quality:** JUnit, Mockito, SonarQube, TDD
- **DevOps & Tools:** Docker, Docker Swarm, Kubernetes, Jenkins, CI/CD, Git, SVN, Maven, Gradle, Jira, Confluence, Bitbucket, Grafana, Flyway, Accusiem, Portainer
- **Cloud:** AWS S3, Azure Blob Storage
- **Practices:** OpenRewrite, Event-Driven Architecture, REST API Versioning, Database Indexing, Performance Profiling, Feature Flags, API Documentation (Swagger/OpenAPI)
- **Frontend (familiarity):** Angular

PROJECTS

SNL — Statiegeld Nederland

Description: National deposit return system in the Netherlands for plastic bottles and metal cans, managing deposit collection, refunds, and logistics for returned containers.

Technologies: Java 21, Spring Boot 4.0, Angular 21, Hibernate ORM, QueryDSL, JPA, WebClient, REST APIs, Elasticsearch, Gradle, Swagger, MySQL, JUnit 5, Mockito, SonarQube, JavaMail, JasperReports, Jenkins, JIRA, GIT, BitBucket, Docker, RabbitMQ, Consul

- Built and maintained scalable backend services for deposit, charity, payout, and simulation modules using Spring Boot and microservices.
- Contributed to migrating core deposit return services to Java 21 and Spring Boot 4.0.
- Implemented dynamic scheduling functionality allowing administrators to configure scheduled jobs without application redeployment.
- Integrated Elasticsearch for search and reporting on high-volume transaction data, and improved scalability by optimizing transaction processing and database access patterns.
- Ensured high code quality and test coverage using JUnit, Mockito, and SonarQube within a Jenkins-managed CI/CD pipeline.
- Created API documentation with Swagger and designed dynamic reports using JasperReports.
- Developed new features based on evolving business requirements.

Reverse Vending Collector (RVC)

Description: Built for Tible, RVC automates deposit collection by integrating with in-store counting machines. It processes transactions from reverse vending machines (e.g., TOMRA) and is deployed across Ecocan Africa, Pure, Plus, Jumbo, Polka, KNSNL and other locations.

Technologies: Java 21, Spring Boot 4.0, Angular 21, Hibernate, QueryDSL, JPA, WebClient, REST APIs, Gradle, Swagger, MySQL, JUnit 5, Mockito, SonarQube, JavaMail, JasperReports, Thymeleaf, Jenkins, JIRA, GIT, BitBucket, Docker, RabbitMQ, Consul

- Designed and maintained backend services for deposit transactions using Spring Boot and microservices for scalable, efficient processing.
- Developed machine approval, discovery, and transaction management modules, integrating reverse vending machines with backend services across multiple international deployments.
- Developed transaction simulation and reporting modules supporting large-scale testing and operational analysis.
- Built a Connector Status web page with Thymeleaf and REST APIs, and designed backend support for connector health monitoring and remote machine management integrated with openresty.
- Implemented secure export and reporting features using JasperReports and cloud storage, with Swagger-documented, authenticated APIs for developer usability and security compliance.
- Designed and developed REST APIs per client requirements to support transaction processing and system integrations.
- Created webhooks to notify client systems of key transaction and status events in real time.
- Integrated third-party APIs into the RVC platform including payment services.
- Developed REST APIs for integration with a Xojo-based client application, implementing OAuth 2.0 client-credentials authentication while maintaining backward compatibility with the existing authentication flow.
- Developed new features based on evolving business requirements.
- Handled client-specific implementations for Ecocan Africa, Pure, Plus, Jumbo, Polka, KNSNL, and other clients each requiring a distinct functional approach based on unique business requirements.

RVC Connector Service

Description: Backend connector service responsible for secure communication between the RVM Collector platform and client-side RVM controllers/machines, enabling remote management, configuration, authentication, and operational monitoring.

Technologies: Java 21, Spring Boot 4.0, Spring Security, WebClient, REST APIs, JWT, Docker, Docker Swarm, Gradle, JUnit, Mockito

- Developed and maintained the RVC Connector Service to enable secure communication between the central RVM Collector platform and remote client controllers.
- Implemented machine authentication and authorization flows using machine tokens and environment-specific validation.
- Developed REST API integrations for machine management, configuration, and communication between the RVC platform and connector services.
- Implemented environment-aware communication supporting TEST, ACCEPT, and PROD deployments with environment-specific service endpoints.
- Developed backend clients using Spring WebClient for secure communication with connector and external services.
- Implemented authentication validation and caching to reduce repeated authentication requests and improve connector communication performance.
- Contributed to remote management of client controllers, including retrieving and updating machine/controller configuration.
- Worked on connector health and operational monitoring, supporting real-time visibility into connector status and machine-related metrics.
- Investigated and resolved connectivity issues across Docker/Docker Swarm environments, including VPN connection and container hostname/environment configuration issues.
- Implemented error handling for external API communication, including HTTP 4xx/5xx responses, timeouts, authentication failures, and environment mismatches.
- Supported production deployments and troubleshooting of connector services across international client environments.
- Developed new features based on evolving business requirements.

OCM — Online Counting Machine

Technologies: Java, Spring Boot, Hibernate/JPA, QueryDSL, MongoDB, REST APIs, Gradle, JUnit, Mockito, SonarQube

- Contributed to the development and maintenance of OCM backend services and enterprise modules.
- Designed and implemented REST APIs based on functional and client requirements.
- Investigated and resolved backend defects while maintaining application stability.
- Refactored existing backend components to improve maintainability, reusability, and code quality.
- Contributed to modernization and dependency upgrades across the application ecosystem.
- Developed new features based on evolving business requirements.

HAWK Platform & Shared Libraries

Description: Developed and maintained a suite of shared Java/Spring Boot libraries used across multiple enterprise applications, providing reusable functionality and common infrastructure for DRS, RVM, financial, article, reporting, and core business services.

Libraries: HAWK Core, HAWK Geo, HAWK Financial, HAWK DRS, HAWK RVM, HAWK Article, HAWK Reports, HAWK Relations

Technologies: Java, Spring Boot, Spring Security, Hibernate/JPA, QueryDSL, REST APIs, Redis, Elasticsearch, Gradle, JUnit, Mockito, SonarQube

- Developed and maintained reusable HAWK libraries consumed by multiple enterprise applications, reducing duplicate implementation and ensuring consistent backend functionality.
- Designed and developed HAWK Geo from scratch as a new shared library, providing reusable geolocation and location-based capabilities for downstream applications.
- Worked on HAWK DRS and HAWK RVM components supporting deposit-return and reverse-vending business functionality.
- Contributed to HAWK Financial, HAWK Article, and HAWK Reports, implementing and maintaining reusable backend functionality across client applications.
- Developed common backend components using Spring Boot, Hibernate/JPA, QueryDSL, and REST APIs for reuse across multiple projects.
- Contributed to Java and Spring Boot version upgrades and modernization of shared libraries while maintaining compatibility with consuming applications.
- Resolved integration, dependency, and compatibility issues across applications consuming HAWK libraries.
- Improved shared library quality through refactoring, automated testing, SonarQube remediation, and maintainability improvements.
- Developed new features based on evolving business requirements.

Menkehorst Nursery

Description: Plant nursery management system supporting seasonal carnival operations, ensuring timely feature updates and system stability.

Technologies: Java 21, Spring Boot 3.5, Hibernate, QueryDSL, JPA, Elasticsearch, Gradle, Swagger, MySQL, SonarQube, JavaMail, Jenkins, JIRA, GIT, BitBucket

- Developed and supported backend services using Spring Boot and microservices to ensure seamless application functionality.
- Delivered feature enhancements aligned with evolving business needs to support carnival operations.
- Collaborated across teams on full-stack improvements, ensuring consistent integration between backend.

Employee Timesheet Management System

Description: Internal timesheet management system, used for employee time tracking, project allocation, and reporting.

Technologies: Java 8, Spring Boot, Hibernate, JPA, MySQL, JUnit

- Developed scalable RESTful APIs to manage timesheet entries, approvals, and project-hours allocation.
- Designed and managed the MySQL database schema, ensuring data integrity and optimal query performance for time-entry records.
- Wrote unit and integration tests using JUnit to validate functionality and maintain high test coverage.
- Refactored and optimized existing code, improving reusability, maintainability, and performance.

PERSONAL PROJECTS

Tracksy – Expense Tracker (Android App, Google Play Store)

Play Store: play.google.com/store/apps/details?id=com.tracksy.app

- Independently designed, built, and published Tracksy, a personal finance and expense-tracking Android app, to the Google Play Store.
- Implemented category-based expense analytics, monthly filtering, CSV export, and biometric app lock, with all data stored locally for user privacy.
- Owned the app end-to-end as a solo developer — architecture, UI/UX, and release to production.

EDUCATION

Bachelor of Technology (B.Tech) in Mechanical Engineering

Jawaharlal College of Engineering and Technology, Palakkad